UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. What happens when an array index is outside its valid range?
   A. Undefined behaviour may occur
   B. C++ automatically resizes the array
   C. The array becomes a string
   D. The compiler always rejects the program

2. What is displayed by the following code?

```
int x = 7;
cout << x / 2 << " " << x % 2;
```

   A. 4 1
   B. 3.5 1
   C. 3 1
   D. 3 0

3. What does a compiler do?
   A. Translates the whole source program before execution
   B. Executes one source statement at a time
   C. Draws flowcharts automatically
   D. Stores only integer values

4. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. n
   B. p
   C. t
   D. i

5. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. iii and iv
   B. i and iii
   C. i and ii
   D. ii and iv

6. Which statement is most suitable for selecting among several constant cases?
   A. for
   B. do...while
   C. while
   D. switch

7. Which identifier is valid in C++?
   A. total_score
   B. 2total

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C. int
D. total score

8. Which loop is guaranteed to execute its body at least once?
A. while
B. for
C. if
D. do...while

9. Which of the following are selection structures?
i. if
ii. switch
iii. for
iv. while
A. ii and iv
B. i and ii
C. i and iii
D. iii and iv

10. Which item is NOT a programming paradigm?
A. Structured programming
B. Debugging
C. Procedural programming
D. Object-oriented programming

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(5 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
    total += i;
}
cout << total;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(9 marks)

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QUESTION 5

A shop stores the item prices shown in Table 1.

a) Declare an array named prices using the values in Table 1. Display all values.

Table 2. Item Prices

Item	Price (RM)
1	20.50
2	55.00
3	15.90
4	75.00
5	10.00

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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